Assignment 8 - BERT Sentiment Analysis

Aim:

Implement a pre-trained BERT model for binary sentiment analysis using the Rotten Tomatoes movie-review dataset.

Concepts

- Pre-trained language model (BERT)
- Transfer learning / fine-tuning
- Tokenization (WordPiece)
- Sequence classification
- Hugging Face Transformers library
- Binary sentiment classification
- Training with Hugging Face Trainer API

Dataset

- Rotten Tomatoes movie reviews (Hugging Face Datasets)
- Classes: Negative (0), Positive (1)
- 2000 training samples
- 500 testing samples
- Balanced class distribution

Tools / Libraries

- Python
- PyTorch
- Hugging Face Transformers (BertForSequenceClassification)
- Hugging Face Datasets
- scikit-learn (accuracy, precision, recall, F1, confusion matrix)
- Matplotlib (visualization)

Model

Pre-trained model: bert-base-uncased

BERT (Bidirectional Encoder Representations from Transformers) is fine-tuned for binary sentiment classification on the Rotten Tomatoes dataset.

- Pre-trained BERT base (110M parameters)
- Fine-tuning with classification head
- 1 epoch of training
- Batch size: 8
- Evaluation during training

Methodology

```
Rotten Tomatoes Dataset
        |
Text Tokenization (BERT WordPiece)
        |
Pre-trained BERT (bert-base-uncased)
        |
Fine-tuning
        |
Sentiment Prediction
        |
Model Evaluation
```

Evaluation Metrics

- Accuracy
- Precision
- Recall
- F1-Score
- Confusion Matrix
- Classification Report

How to Run

- Install required libraries: transformers, datasets, scikit-learn, torch
- Run the notebook / script
- The Rotten Tomatoes dataset will be loaded automatically
- BERT will be fine-tuned on the training data
- The model will be evaluated on the test data
- New movie reviews will be classified as Positive or Negative

Example

Input:

```
This movie was absolutely fantastic.
```

Output:

```
Sentiment: Positive
```

Conclusion

The project demonstrates how a pre-trained BERT model can be fine-tuned for sentiment analysis and used to classify movie reviews into positive and negative sentiments.